

Classification Lecture 7

Thursday, 3 October 2024

6:13 AM

H/W

Candidate	Skill Level	Certification	Previous Job Success	Education Relevance	Hired
1	High	Yes	High	Relevant	Yes
2	High	No	Low	Irrelevant	No
3	Medium	Yes	Medium	Relevant	Yes
4	Low	No	Low	Irrelevant	No
5	Medium	No	Medium	Irrelevant	No
6	Low	Yes	High	Relevant	No
7	High	Yes	Medium	Irrelevant	Yes
8	Medium	Yes	Low	Relevant	No
9	High	No	High	Relevant	Yes
10	Low	No	Medium	Irrelevant	No

Classify whether a new candidate with skill level: Medium, has done certification, Previous Job Success: High, Education Relevant as hired or not?

Sol. $\vec{x} = (M, Y, H, R)$

$$P(Y | M, Y, H, R) = \frac{P(M, Y, H, R | Y) \cdot P(Y)}{P(M, Y, H, R)} \propto P(M, Y, H, R | Y) \cdot P(Y)$$

$$= P(M | Y) \cdot P(Y | Y) \cdot P(H | Y) \cdot P(R | Y) \cdot P(Y)$$

$$= \frac{1}{4} \times \frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} \times \frac{4}{10}$$

$$= \frac{18}{640} \approx 0.028$$

$$P(N | M, Y, H, R) \propto P(M | N) \cdot P(Y | N) \cdot P(H | N) \cdot P(R | N) \cdot P(N)$$

$$= \frac{2}{6} \times \frac{2}{6} \times \frac{1}{6} \times \frac{2}{6} \times \frac{6}{10}$$

$$= \frac{8}{2160} \approx 0.0037$$

$$P(Y | \dots) > P(N | \dots) \Rightarrow \boxed{\text{Hired}}$$

Laplace Smoothing

Candidate	Skill Level	Certification	Previous Job Success	Education Relevance	Hired
1	High	Yes	High	Relevant	Yes ←
2	High	No	Low	Irrelevant	No
3	Medium ^{low?}	Yes	Medium	Relevant	Yes ←
4	Low	No	Low	Irrelevant	No
5	Medium	No	Medium	Irrelevant	No
6	Low	Yes	High → Medium	Relevant	No
7	High	Yes	Medium	Irrelevant	Yes ←
8	Medium	Yes	Low	Relevant	No
9	High	No	High	Relevant	Yes ←
10	Low	No	Medium	Irrelevant	No

$$P(Y|M, Y, H, R) \propto \dots = 0, \quad P(N|M, Y, H, R) \propto \dots = 0$$

Zero probability effect

without Laplace smoothing :-

$$P(\text{feature} | \text{class}) = \frac{\# \text{ times feature appears in class}}{\# \text{ Total instances in class}}$$

With Laplace Smoothing:-

$$P(\text{feature} | \text{class}) = \frac{\# \text{ times feature appears in class} + 1}{\# \text{ Total instances in class} + m}$$

α
 $\alpha \cdot m$

m : # different values this feature can take

(A/B/C) (P/Q)

x_1	x_2	y
A.	P.	0 ←
B	P	1
C	Q	0 ←
A.	P.	0 ←

$$\left\{ \begin{aligned} P(x_1=B | y=0) &= \frac{0+1}{3+3} = \frac{1}{6} \\ P(x_1=A | y=0) &= \frac{2+1}{3+3} = \frac{3}{6} \\ P(x_1=C | y=0) &= \frac{1+1}{3+3} = \frac{2}{6} \end{aligned} \right.$$

$$P(x_2=P | y=0) = \frac{2+1}{3+2} = \frac{3}{5}$$

$$P(x_2=Q | y=0) = \frac{1+1}{3+2} = \frac{2}{5}$$

$$x = B, Q$$

$$P(Y=0 | B, Q) \propto P(B|0) \cdot P(Q|0) \cdot P(0) \dots$$

$$= \frac{1}{6} \times \frac{2}{5} \times \frac{3}{4} = \frac{1}{20}$$

How to handle numerical data

In Naive Bayes?

Numerical		Categorical		
age	Salary		Gender	
x_1	x_2 ...		x_p	$y(0/1)$
25	21000		M	
25	25000		M	
...	...		...	
17	10000		F	
...	...		...	

$$x = (30, 35000, \dots, F)$$

$$P(x_1 = 30 | y = -) \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Fit a distribution}$$

$$P(x_2 = 35000 | y = -)$$

$$P(x_p = F | y = -) \checkmark$$

$$P(x_1 | y = 0) \rightarrow$$

$$P(x_2 | y = 0) \rightarrow$$

$$P(x_1 | y = 1) \rightarrow$$

$$P(x_2 | y = 1) \rightarrow$$

Fit a distribution
& calculate the
parameters

Gaussian Naive Bayes

eg.	Numerical Age (x_1)	Numerical Salary (x_2)	Categorical Education (x_3)	Buy (c)
	25	50,000	Bachelors	Not Buy
	<u>30</u>	80,000	Masters	<u>Buy</u>
	35	65,000	High School	Not Buy
	<u>40</u>	100,000	Bachelors	<u>Buy</u>

$x = (32, 75000, \text{Bachelors})$, predict the label Buy using Naive Bayes. (Gaussian Naive Bayes)

Sol. Can we apply Laplace smoothing? No.

We assume normal distributions Age, Buy $\sim N(\mu_{A,B}, \sigma_{A,B}^2)$

Age, Buy $\sim N(\mu_{A,B}, \sigma_{A,B}^2)$

Age, NB $\sim N(\mu_{A,NB}, \sigma_{A,NB}^2)$

Salary, B $\sim N(\mu_{S,B}, \sigma_{S,B}^2)$

$$\mu_{A,B} = \frac{30+40}{2} = 35$$

$\underbrace{\quad}_{\{30,40\}}$

$$\sigma_{A,B} = \sqrt{\frac{(30-35)^2 + (40-35)^2}{2}} = 5$$

$$A, B \sim N(35, 25)$$

$$P(A=32|B) = \frac{1}{\sigma_{A,B} \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu_{A,B}}{\sigma_{A,B}} \right)^2}$$

likelihood

$$= \frac{1}{5 \times \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{32-35}{5} \right)^2} \approx \underline{0.067}$$

$$P(B | A=32, S=75000, E=\text{Bachelors})$$

$$\propto \underbrace{P(A=32|B)}_{\downarrow} \cdot \underbrace{P(S=75000|B)}_{\downarrow} \cdot \underbrace{P(E=\text{Bachelors}|B)}_{\downarrow} \cdot P(B)$$

$$\approx 0.067 \times \text{---} \times \left(\frac{1}{2}\right)$$

$$\approx \dots \quad \downarrow \quad \left(\frac{2}{5}\right) \text{ if we use Laplace}$$

ex

Bernoulli Naive Bayes

eg. #	Tweet Id	Contains "Happy" (0/1)	Contains "Sad" (0/1)	Positive Sentiment y
	1	1	0	1 ←
	2	0	1	0 ←
	3	1	1	1 ←

$$x = (1, 0)$$

classify using
Bernoulli Naive
Bayes

Sol. Happy, $y=1 \sim \text{Bernoulli}(p) \quad \therefore p = \frac{2}{2} = 1$
 $\downarrow$
 $\{1, 1\}$

Sad, $y=1 \sim \text{Bernoulli}(p) \quad \therefore p = \frac{1}{2}$

$$P(1, 0 | y=1) = P(\text{Happy}=1 | y=1) \cdot P(\text{Sad}=0 | y=1)$$

likelihood

$$= 1 \times (1 - \frac{1}{2}) = 1 \times \frac{1}{2} = \frac{1}{2}$$

∴

-x-

Multinomial Naive Bayes

$$P(X_1=x_1, X_2=x_2, \dots, X_p=x_p \mid Y=c) \propto \theta_{1,c}^{x_1} \cdot \theta_{2,c}^{x_2} \cdot \theta_{3,c}^{x_3} \cdots \theta_{p,c}^{x_p}$$

$$= \frac{n!}{x_1! x_2! \cdots x_p!} \cdot \theta_{1,c}^{x_1} \theta_{2,c}^{x_2} \cdots$$

eg.

Article Id	x_1 "win"	x_2 "game"	x_3 "computer"	x_4 "election"	y category
1	3	2	0	0	Sports ←
2	0	0	2	1	Technology ←
3	0	1	1	2	Politics ←
4	1	2	0	0	Sports ←
5	0	0	3	0	Technology ←

$x = (1, 0, 1, 1)$ classify the point using Multinomial Naive Bayes
(use laplace smoothing)

Sol. $P(x=1,0,1,1 \mid y=\text{Sports})$

$$\propto (\theta_{x_1, \text{Sports}})^1 \cdot (\theta_{x_2, \text{Sports}})^0 \cdot (\theta_{x_3, \text{Sports}})^1 \cdot (\theta_{x_4, \text{Sports}})^1$$

$$\theta_{x_i, y=c} = \frac{\text{Count}(x_i \mid y=c)}{\sum_c \text{Count}(x_i \mid y=c)} \quad \left(= \frac{\text{Count}(x_i \mid y=c) + 1}{\sum_c [\text{Count}(x_i \mid y=c) + 1]} \right)$$

$$\theta_{x_1, \text{Sports}} = \frac{4+1}{8+4} = \frac{5}{12}, \quad \theta_{x_2, \text{Sports}} = \frac{4+1}{8+4} = \frac{5}{12}, \quad \theta_{x_3, \text{Sports}} = \frac{0+1}{8+4} = \frac{1}{12}$$

$$\theta_{x_4, \text{Sports}} = \frac{0+1}{8+4} = \frac{1}{12}$$

$$P(x=1,0,1,1 \mid y=\text{Sports}) \propto \left(\frac{5}{12}\right)^1 \cdot \left(\frac{5}{12}\right)^0 \cdot \left(\frac{1}{12}\right)^1 \cdot \left(\frac{1}{12}\right)^1 = \frac{5}{(12)^3}$$

$$P(y=\text{sports} | x=1,0,1,1) \propto P(x=1,0,1,1 | y=\text{sports}) \cdot P(y=\text{sports})$$

$$= \frac{\cancel{5}}{(12)^3} \cdot \frac{2}{\cancel{5}} = \frac{2}{(12)^3} \approx 0.0012$$

$$\theta_{x_1, \text{Tech}} = \frac{0+1}{6+4} = \frac{1}{10}, \quad \theta_{x_2, \text{Tech}} = \frac{0+1}{6+4} = \frac{1}{10}, \quad \theta_{x_3, \text{Tech}} = \frac{5+1}{6+4} = \frac{6}{10},$$

$$\theta_{x_4, \text{Tech}} = \frac{1+1}{6+4} = \frac{2}{10}$$

$$P(y = \text{Tech} | x=1,0,1,1) \propto P(x=1,0,1,1 | y=\text{Tech}) \cdot P(y=\text{Tech})$$

$$\propto \left(\frac{1}{10}\right)^1 \cdot \left(\frac{1}{10}\right)^0 \cdot \left(\frac{6}{10}\right)^1 \cdot \left(\frac{2}{10}\right)^1 \cdot \frac{2}{5}$$

$$= \frac{24}{5000} = 0.0048$$

$$\theta_{x_1, \text{politics}} = \frac{0+1}{4+4} = \frac{1}{8}, \quad \theta_{x_2, \text{politics}} = \frac{1+1}{4+4} = \frac{2}{8}, \quad \theta_{x_3, \text{politics}} = \frac{1+1}{4+4} = \frac{2}{8}$$

$$\theta_{x_4, \text{politics}} = \frac{2+1}{4+4} = \frac{3}{8}$$

$$P(y = \text{politics} | x = 1, 0, 1, 1) \propto \left(\frac{1}{8}\right)^1 \cdot \left(\frac{2}{8}\right)^0 \cdot \left(\frac{2}{8}\right)^1 \cdot \left(\frac{3}{8}\right)^1 \cdot \frac{1}{5}$$

$$= \frac{6}{8^3 \times 5} \approx 0.0023$$

$\therefore P(y = \text{Technology} | \dots)$ is highest $\therefore$ Technology

x